VIJAYABHARATHI R

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Summary

Computer Science Engineering graduate with a strong foundation in Java, Data Structures & Algorithms, OOP, and Full Stack Development. Skilled in Spring Boot, REST APIs, and MySQL with hands-on experience through internship and academic projects. Seeking a Software Engineer role to apply problem-solving, backend development, and scalable application development skills.

Education

B.E. – Computer Science and Engineering *Nov 2022 – Apr 2026*
University College of Engineering BIT Campus, Tiruchirappalli

- CGPA: 8.12/10

HSC *Nirmala Higher Secondary School, Chidambaram* *2020 – 2022*

- Percentage: 91.6%

SSLC *Sri Nataraja Matriculation School, Chidambaram* *2019 – 2020*

- Percentage: 93.8%

Technical Skills

Languages: Java (Intermediate), Python, C, SQL

Frameworks: Spring Boot, Spring Data JPA, Hibernate, Maven

Database: MySQL

Concepts: MVC Architecture, REST APIs, JWT Authentication, RBAC, OOP, Data Structures & Algorithms, Machine Learning (Regression, ARIMA, Random Forest, KNN, Reinforcement Learning)

Developer Tools: IntelliJ IDEA, VS Code

Experience

Java Full Stack Developer Intern *iLife Technologies, Trichy* *Jul 2025 – Aug 2025*

- Built an **Employee Management System** (Java Spring Boot, Spring Data JPA, MySQL) with full CRUD operations and MVC architecture, streamlining employee record management for the organisation.
- Designed the relational DB schema with indexed lookups and applied **ORM-based JPA** queries to reduce data retrieval overhead, gaining practical full-stack and database design experience.

Projects

AI-Based Hospital Patient Flow & Resource Management System *Tools: Java 21, Spring Boot 3.3.5, MySQL 8, Maven, IntelliJ IDEA*

- Engineered a full-stack hospital management system integrating **4 ML algorithms** (KNN, Random Forest, Reinforcement Learning, Genetic Algorithm) for real-time patient triage, bed allocation, and queue scheduling – reducing simulated bed allocation time by **40%** vs manual assignment.
- Implemented **JWT-based RBAC** for 4 user roles with live dashboards and AI-driven ambulance severity scoring, securing all REST API endpoints end-to-end.

Dynamic Hotel Pricing using Machine Learning *Tools: Python, Scikit-learn, Pandas, Jupyter Notebook*

- Trained **Linear Regression and Random Forest** models on **10,000+ booking records** across demand, seasonality, and competitor pricing features – achieving **12% RMSE reduction** over the baseline.
- Applied **ARIMA time-series forecasting** for 7-day occupancy prediction, driving data-driven rate decisions that improved simulated hotel revenue by **18%** against a fixed-rate strategy.

Achievements

- Scored **91.6% in HSC** and **93.8% in SSLC**, demonstrating consistent academic excellence.
- Completed internship (1 month) delivering a production-ready EMS using **Spring Boot, JPA, MVC, MySQL** independently.
- Served as **Company Quarter Master Sergeant (CQMS)** in the **National Cadet Corps (NCC)** – led logistics and resource management for a cadet company, demonstrating strong **leadership, discipline, and team coordination** skills.
- Strong core foundations in **Data Structures & Algorithms, OOP, and DBMS** through graduate coursework.